FOOTPRINTING & RECONNAISSANCE REPORT WITH theHARVESTER

W2-PM4 | WEEK 2 – PROJECT MODULE 4
CYBERSECURITY & ETHICAL HACKING

Project	Footprinting & Reconnaissance with theHarvester (passive OSINT)
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Program	Networkwalks | Week 2, Project Module 4
Report date	12 November 2026
Environment	Kali Linux (VirtualBox virtual machine)
Tool version	theHarvester 4.11.1
Target domain	microsoft.com (domain specified in the lab guide)

Prepared for educational and portfolio purposes. Findings are limited to the screenshots and terminal outputs captured for this project.

1. Liability Disclaimer

This activity was conducted for cybersecurity education and portfolio documentation, following the Networkwalks lab guide. theHarvester only reads publicly available sources such as search engines and does not send traffic to the target itself, but reconnaissance must still only be performed within an authorized scope. Do not use this technique against organizations without permission. The observations below are point-in-time results and do not establish that any system is vulnerable.

2. Introduction

This report documents W2-PM4: Footprinting and Reconnaissance with theHarvester. theHarvester is a Python tool, pre-installed in Kali Linux, that gathers e-mail addresses, sub-domains, hosts and employee names from public sources. Footprinting is the first stage of an assessment: it shows what an attacker could learn about an organization before touching its systems.

3. Objectives

- Run theHarvester against microsoft.com using the Baidu source with a result limit of 1000 (Task 1).
- Run theHarvester against microsoft.com using all sources with a result limit of 50 (Task 2).
- Identify the e-mail addresses and sub-domains returned by each source.
- Save the output of each run and keep screenshot evidence.

4. Scope and Methodology

In scope: passive collection of public information about the domain microsoft.com, as specified in the lab guide.

Tool and platform: theHarvester 4.11.1 on Kali Linux running in a VirtualBox virtual machine.

Method: the help screen was reviewed to confirm the supported sources, the two tasks were run from the terminal, and the results were saved with the `-f` option (XML and JSON). The saved files were then listed and checked.

Limitations: the result of a run depends on the data source at that moment and can differ between runs; the all-sources command could not be completed (Section 6.3); the terminal output was not saved to a text file; and only three screenshots were captured (Section 11).

5. Tools Used

Tool	Purpose
theHarvester 4.11.1	OSINT tool that collects hosts, sub-domains, e-mails and names from public sources.
Kali Linux terminal	Environment used to run theHarvester and inspect the saved files.
Oracle VirtualBox	Virtualization platform hosting the Kali Linux machine.
Baidu and DuckDuckGo sources	Search-engine data sources queried by theHarvester (`-b baidu`, `-b duckduckgo`).

6. Activities Performed and Results

6.1 Help Output and Tool Version

Running `theHarvester -h` (Figure 1) confirmed version **4.11.1**. Relevant options:

Option	Meaning
`-d, --domain`	Company name or domain to search.
`-l, --limit`	Limit the number of search results (default 500).
`-b, --source`	Select the data source.
`-f, --filename`	Save the results to an XML and a JSON file.

The source list shown by the help screen includes baidu, brave, duckduckgo, crtsh, hunter, virustotal and yahoo, among others. It does not list an `all` source.

6.2 Task 1 – Baidu Search (limit 1000)

```
theHarvester -d microsoft.com -l 1000 -b baidu
```

The terminal output in Figure 2 shows: Target microsoft.com, “Searching Baidu”, and **no IPs, no e-mails, no people and no hosts found**. The same command repeated with `-f task1-baidu` (Section 6.5) also produced no hosts.

Earlier 9-host transcript. A pasted terminal transcript from an earlier Baidu run listed 9 hosts (account, c.urs, developer, dotnet, learn, o15.officeredir, prod.support.services, support and ti.defender, all under microsoft.com). No screenshot or saved file supports that run, so it is recorded here as an unverified observation and is not used for the results. This variation is expected because search-engine sources change between runs.

6.3 Task 2 – All-Sources Search (limit 50)

The lab guide specifies:

```
theHarvester -d microsoft.com -l 50 -b all
```

The command captured in Figure 3 was typed as `-b all~` (with a trailing tilde, a typing error). theHarvester answered “The following engines are not supported: {'all~'}” and “[!] Invalid source.” This error comes from the typo, so it does not by itself prove that `all` is unsupported. The help screen of this version does not list `all`, so the command should be re-run exactly as written (without the tilde) and the result recorded. Until then, the all-sources search is reported as **not completed**.

6.4 Task 2 – DuckDuckGo Search (alternative source)

Because the all-sources run was not completed, DuckDuckGo was used as an alternative source:

```
theHarvester -d microsoft.com -l 50 -b duckduckgo
```

The run reported **27 hosts** and no IPs, e-mails or people. The raw output was:

2Fblogs.microsoft.com	azuremarketplace.microsoft.com	go.microsoft.com
2Fdocs.microsoft.com	blogs.microsoft.com	learn.microsoft.com
2Fdownload.microsoft.com	browserdefaults.microsoft.com	msdn.microsoft.com
2Fmsdn.microsoft.com	careers.microsoft.com	news.microsoft.com
2Fnews.microsoft.com	choice.microsoft.com	privacy.microsoft.com
Docs.microsoft.com	copilot.microsoft.com	schemas.microsoft.com
account.microsoft.com	developer.microsoft.com	support.microsoft.com
appsource.microsoft.com	docs.microsoft.com	techcommunity.microsoft.com
azure.microsoft.com	download.microsoft.com	visualstudio.microsoft.com

Data-quality note. Five entries begin with “2F” (2Fblogs, 2Fdocs, 2Fdownload, 2Fmsdn, 2Fnews). “2F” is the URL-encoded form of “/”, so these are parsing artifacts of the search results and duplicate blogs, docs, download, msdn and news. “Docs.microsoft.com” repeats docs.microsoft.com in different capitalization. After removing these six, **21 unique sub-domains** remain.

6.5 Exporting Results to XML and JSON

A working folder was created and both scans were run with the `-f` option:

```
mkdir -p ~/W2-PM4
cd ~/W2-PM4
theHarvester -d microsoft.com -l 1000 -b baidu -f task1-baidu
theHarvester -d microsoft.com -l 50 -b duckduckgo -f task2-duckduckgo
```

Both runs displayed “Reporting started”, “XML File saved” and “JSON File saved”. The `ls -lh` listing confirmed four files:

```
total 16K
-rw-rw-r-- 1 kali kali 81 Nov 12 14:33 task1-baidu.json
-rw-rw-r-- 1 kali kali 126 Nov 12 14:33 task1-baidu.xml
-rw-rw-r-- 1 kali kali 745 Nov 12 14:34 task2-duckduckgo.json
-rw-rw-r-- 1 kali kali 1.1K Nov 12 14:34 task2-duckduckgo.xml
```

6.6 Contents of the Saved Files

File	Content
`task1-baidu.json`	Command string, `"hosts":[]` and `"shodan":[]` (81 bytes).
`task1-baidu.xml`	Only the <cmd> element; no host elements (126 bytes).
`task2-duckduckgo.json`	Command string, the 27 hosts listed in Section 6.4 and an empty shodan array (745 bytes).
`task2-duckduckgo.xml`	Command element and 27 <host> elements (1.1K).

Consistency check. A JSON file built from the command string and the 27 hosts above is exactly 745 bytes, and the matching XML file is 1,061 bytes (shown as 1.1K by ls -lh). Both match the `ls -lh` listing, which confirms that the saved files contain the 27 hosts of the DuckDuckGo run. The 81-byte Baidu JSON file likewise matches an empty hosts array.

7. Summary of Findings

Run	Source / limit	Hosts	E-mails	Evidence
Task 1	Baidu / 1000	0	0	Figure 2; task1-baidu.json and .xml
Task 2 (all sources)	all / 50	–	–	Not completed; typo in command (Figure 3)
Task 2 (alternative)	DuckDuckGo / 50	27 raw, 21 unique	0	Terminal output; task2-duckduckgo.json and .xml

Neither source returned an e-mail address or a person's name, although both tasks ask for e-mails and sub-domains. This does not mean none exist: each source covers only part of the public data, and the lab guide recommends trying more than one source. Baidu returned nothing in the captured runs, while DuckDuckGo returned a useful set of sub-domains such as account, developer, support, learn, azure, careers and copilot.

8. Risk Analysis / Impact

Observation	Potential impact	Priority
21 unique microsoft.com sub-domains discovered from one public source	Each sub-domain is a possible entry point and shows an attacker which services exist.	Informational
Search results contain malformed or duplicate entries (2F prefix, case differences)	Raw output overstates the number of hosts; results must be cleaned and validated before use.	Data quality
No e-mail addresses returned by Baidu or DuckDuckGo	Low e-mail exposure through these sources only; other sources may expose e-mails used for phishing.	Informational
Results differ between runs and sources	A single run is not a complete picture; several sources are needed for coverage.	Evidence gap

9. Security Recommendations

- Maintain an inventory of all public sub-domains and remove DNS records for unused ones.
- Run theHarvester against your own domain regularly to see what an attacker would see.
- Limit public exposure of staff e-mail addresses; use role-based addresses and e-mail filtering.
- Train staff to recognize phishing e-mails that use harvested addresses.
- Query several sources (for example crtsh, hunter, virustotal) to get fuller coverage.
- Save every run to a text file (`| tee output.txt`) and keep screenshots as evidence.

10. Conclusion

The W2-PM4 exercise demonstrated passive footprinting with theHarvester 4.11.1 on Kali Linux. The Baidu search of microsoft.com (limit 1000) returned no hosts or e-mails in the captured runs. The all-sources search (limit 50) could not be completed because the command was entered with a trailing tilde; it should be repeated as written in the lab guide. As an alternative, a DuckDuckGo search returned 27 raw hosts, which reduce to 21 unique sub-domains after removing artifacts. The results were saved in XML and JSON files and checked against the file listing.

Remaining items: (1) re-run `theHarvester -d microsoft.com -l 50 -b all` without the tilde and capture the screenshot; (2) save the output of each task to a text file, as the lab guide requires; (3) add screenshots of the DuckDuckGo run, the `ls -lh` listing and the file contents.

11. Evidence Register

Item	Status / evidence
Help screen and version check	Completed – Figure 1 (version 4.11.1).
Task 1 – Baidu search	Completed – Figure 2 and saved JSON/XML files; 0 hosts, 0 e-mails.
Task 2 – all sources	Not completed – Figure 3 shows the typo (`all~`); re-run required.
Task 2 – DuckDuckGo (alternative)	Completed – 27 raw / 21 unique hosts; terminal text and saved files, no screenshot.
XML and JSON export	Completed – four files shown by `ls -lh` (terminal text, no screenshot).
Text-file output required by the guide	Not captured.

12. References

1. Networkwalks, *W2-PM4 – Week 2 Project Module 4: Footprinting & Reconnaissance with theHarvester* (practical guide).

2. theHarvester project: https://github.com/laramies/theHarvester

3. Kali Linux tools: https://www.kali.org/tools/theharvester/

Appendix – Selected Screenshots

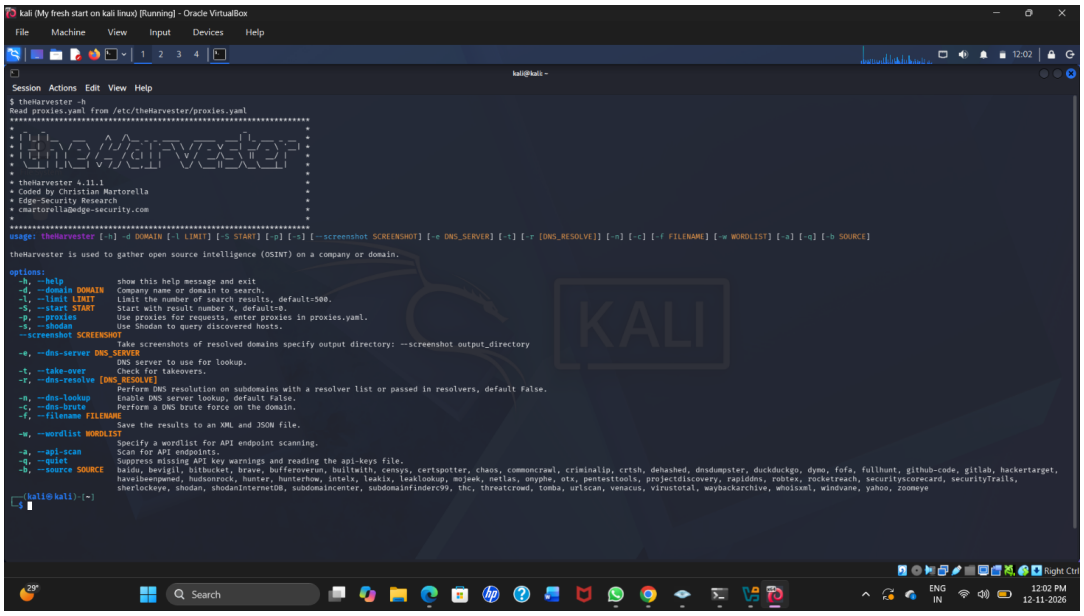


Figure 1. theHarvester help screen showing version 4.11.1 and the supported sources.



Figure 2. Task 1 – Baidu search (limit 1000): no hosts found.



Figure 3. Task 2 – command typed as -b all~ rejected as an invalid source.